

Computer Languages Python

COSC-2110

FA 2020 Section 01 3 Credits 08/24/2020 to 12/11/2020 Modified 06/27/2020

Meeting Times

Lecture

Monday, Wednesday, Friday, 1:00 PM to 1:50 PM, EAB108

Office Hour

Monday, Wednesday, Thursday, Friday, 2:00 PM to 4:00 PM, EAB 371

Contact Information

Instructor: Dr. Xiaoyuan Suo

Email: xiaoyuansuo51@webster.edu

Office: EAB 371

Phone: 314-246-2359

Description

Investigates different computer languages, offered under different subtitles. May be repeated for credit if content differs.

Requisites

None

Outcomes

This course covers programming and problem solving using Python. Topics include procedures and functions, iteration, recursion, sequence, dictionaries, arrays and vectors, strings, algorithms, object-oriented programming, and GUIs (graphical user interfaces).

1. Write moderately complex Python applications.
2. Use various control structures and to use functions.
3. Utilize the concepts of event-driven programming, object-oriented programming.
4. Develop graphical user interfaces and sophisticated operational programs.
5. Identify parallelizable problem and utilize parallel Python

Required Textbook

Textbook varies by topic

Deliverables

✓ Evaluation

Criteria

Exam 1 - 3	45%	15% each
Exam 4 (Final Exam)	20%	Final exam is comprehensive
Assignment (12 programming assignment)	25%	All programs must be submitted and run correctly to pass the course All programs must execute according to specifications If an assignment is submitted late, 20% is deducted from the grade, with an additional 2% per day deduction, thereafter.
Attendance (Homework and participation)	10%	Attendance and participation are required. Students who cannot attend every class are advised to withdraw from the course. Please notify the instructor as soon as possible regarding an excused absence. A student's final grade will be lowered for each unexcused absence. From time to time there will be homework (1 or 2 questions) assigned, your submission will count toward the attendance grade.

lowest % for an A is 93%, lowest % for an A- is 90%, lowest % for an B+ is 87%, lowest % for an B is 83%, lowest % for an B- is 80%, lowest % for an C+ is 77%, lowest % for an C is 73%, lowest % for an C- is 70%, lowest % for an D+ is 67%, lowest % for an D is 63%, any lower grade earns a Fail

There will be 6 Chapter tests.

The UNDERGRADUATE catalog provides these guidelines and grading options: A, A- superior work in the opinion of the instructor

B+, B, B- good work in the opinion of the instructor

C+, C, C- satisfactory work in the opinion of the instructor

D+, D passing, but less than satisfactory work in the opinion of the instructor I incomplete work in the opinion of the instructor

ZF An incomplete which was not completed within one year of the end of the course F unsatisfactory work in the opinion of the instructor; no credit is granted

W withdrawn from the course IP course in progress NR not reported for the course

Z a temporary designation given by the registrar indicating that the final grade has not been submitted by the instructor. When the final grade is filed in the Office of the Registrar, that grade will replace the Z.

Schedule

Week 1	The Python programming language • Development environment • Compiling a simple Python Application • Data types • Input/Output: Print function Reading: Chapters 1, 2
Week 2	Topics • The if/else statement • Loop Reading: Chapters 3, 4
Week 3	Topics • Defining and calling a function • File processing Reading: Chapters 5, 6
Week 4	Topics • List, dictionary, and sets Reading: Chapters 7, 9 Exam: Friday, topics from Chapter1 - 7 and 9
Week 5	Topics • Class and objects Reading: Chapter 10, 11
Week 6	Topics • Recursion Reading: Chapter 12
Week 7	Topics • GUI Reading: Chapter 13

Week 8	Topics • GUI continued Reading: Chapter 13 Exam: Friday, topics from Chapter 10-13
Week 9	Topics • Why parallel programming? • Identifying parallel programming problems Reading: Handout on contextualizing parallel, concurrent, and distributed programming
Week 10	Topics • Design parallel algorithms ◦ divide and conquer ◦ decomposing data Reading: Handout on parallel algorithms
Week 11	Topics • Using threading ◦ Fibonacci series with multiple inputs Reading: Handout on parallel algorithms
Week 12	Topics • Using multiprocessing ◦ Concepts and discussions • Interprocess communication Reading: Handout on multiprocessing, in class projects on using multiprocessing to compute Fibonacci series terms with multiple inputs Exam: Friday, topics on Parallel Python
Week 13	Topics • Python database ◦ opening and closing a database connection ◦ database fundamentals Reading: Chapter 14

Week 14	Topics - Python database - creating a table - adding data to a table - select and update Reading: Chapter 14
Week 15	Topics More on Python database Reading: Chapter 14
Week 16	Final exam, Comprehensive

* Course Policies

- Attendance and participation are required. Students who cannot attend every class are advised to withdraw from the course. Please notify the instructor as soon as possible regarding an excused absence. A student's final grade will be lowered for each unexcused absence.
- No cell phone usage during class time, including sending or receiving of phone calls, text messages, or instant messages. Phones should be turned off, not set to vibrate, etc. Phones should be not on the desktop. Please do not leave the classroom between class time to get a snack, to get a drink, to use the restroom, etc., unless it is a true emergency.
- Students are responsible for making arrangements with other students for missed class notes and with the instructor for handouts and assignments. Students should make arrangements with the instructor beforehand for not being able to attend any test or exam. Any unexcused absence will result in a failing grade for that quiz or exam.
- All program assignments must meet the specified style guidelines.
- All program assignments must compile and run (according to specifications) upon submission.
- All program assignments must be successfully completed to pass the course. For all assignment, please follow all written as well as verbal instructions given in class. Students are encouraged to discuss concepts and algorithms of the projects together but must code each program independently.
- Academic Dishonesty: Submitting an assignment that was copied, whole or in part, from another student (past or present) constitutes a violation of student conduct and will be handled according to the guideline of the Student Handbook.
- Facilitating Academic Dishonesty: Assisting any person commit an act of academic dishonesty also constitutes a violation of student conduct and will be handled according to the guideline of the Student Handbook.
- Assignments are due at due date and time indicated. Late assignments will get lower grades. If an assignment is submitted after a deadline, 20% will be deducted from the grade, with an additional 2% per day deduction thereafter.
- Quizzes, Tests and Exams are the property of the Math and Computer Science Department.
- This syllabus may be revised at the discretion of the instructor without the prior notification or consent of the student.

Institutional Policies

Academic Policies

Academic policies provide students with important rights and responsibilities. Students are expected to familiarize themselves with all academic policies that apply to them. Academic policies for undergraduate students can be found in the Undergraduate Studies Catalog; graduate students should review the Graduate Studies Catalog.

Undergraduate Studies Catalog

The Undergraduate Studies Catalog contains academic policies that apply to all undergraduate students. The *academic policies and information* section of the catalog contains important information related to **attendance, conduct, academic honesty, grades, and more**. If you are an undergraduate student, please review the catalog each academic year. The current Undergraduate Studies Catalog is at:

<http://www.webster.edu/catalog/current/undergraduate-catalog/> (<http://www.webster.edu/catalog/current/undergraduate-catalog/>)

Graduate Studies Catalog

The Graduate Studies Catalog contains academic policies that apply to all graduate students. The *academic policies* section of the catalog contains important information related to **conduct, academic honesty, grades, and more**. If you are a graduate student, please review the catalog each academic year. The current Graduate Studies Catalog is at:

<http://www.webster.edu/catalog/current/graduate-catalog/> (<http://www.webster.edu/catalog/current/graduate-catalog/>)

Grading

The Grades section of the academic catalog outlines the various grading systems courses may use, including the information about the final grade reported for this class.

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/academic-policies.html#grading>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/academic-policies.html#grading>)

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/academic-policies.html#grades>
(<http://www.webster.edu/catalog/current/graduate-catalog/academic-policies.html#grades>)

Incomplete

There are important policies that govern grades of Incomplete (I), including the circumstances under which Incomplete grades are granted, deadlines for completion, and consequences should the remaining course work not be completed. It is the responsibility of a student who requests an Incomplete to ensure that he/she understands and follows the policies.

Grade Appeals

Instructors are responsible for assigning grades, and student should discuss grade issues with the instructor. Policies and procedures for appealing grades are available in the appropriate catalog.

Academic Honesty Policy

Webster University is committed to academic excellence. As part of our Statement of Ethics, we strive to preserve academic honor and integrity by repudiating all forms of academic and intellectual dishonesty, including cheating, plagiarism and all other forms of academic dishonesty. Academic dishonesty is unacceptable and is subject to a disciplinary response. Students are encouraged to talk to instructors about any questions they may have regarding how to properly credit others' work, including paraphrasing, quoting, and citation formatting. The university reserves the right to utilize electronic databases, such as Turnitin.com, to assist faculty and students with their academic work.

The University's Academic Honesty Policy is published in academic catalogs:

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/academic-policies.html>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/academic-policies.html>)

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/academic-policies.html>
(<http://www.webster.edu/catalog/current/graduate-catalog/academic-policies.html>)

As a part of the University commitment to academic excellence, the Academic Resource Center provides student resources to become better acquainted with academic honesty and the tools to prevent plagiarism in its many forms:

http://www.webster.edu/arc/plagiarism_prevention/ (http://www.webster.edu/arc/plagiarism_prevention/)

Statement of Ethics

Webster University strives to be a center of academic excellence. The University makes every effort to ensure the following:

- The opportunity for students to learn and inquire freely
- The protection of intellectual freedom and the rights of professors to teach
- The advancement of knowledge through scholarly pursuits and relevant dialogue

To review Webster University's statement of ethics, see the Undergraduate Studies Catalog and the Graduate and Studies Catalog:

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/overview.html#ethics>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/overview.html#ethics>)

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/ethics.html> (<http://www.webster.edu/catalog/current/graduate-catalog/ethics.html>)

Contact Hours for this Course

It is essential that all classes meet for the full instructional time as scheduled. A class cannot be shortened in length. If a class session is cancelled for any reason, the content must be covered at another time.

Important Academic Resources

Academic Accommodations

Webster University makes every effort to accommodate individuals with academic/learning, health, physical and psychological disabilities. To obtain accommodations, students must identify themselves and provide documentation from a qualified professional or agency to the appropriate campus designee or the Academic ADA Coordinator at the main campus. The Academic ADA Coordinator may be reached at 314-246-7700 or disability@webster.edu (<mailto:disability@webster.edu>).

If you have already identified as a student with a documented disability and are entitled to classroom or testing accommodations, please inform the instructor of the accommodations you will require for this class at the beginning of the course.

Academic Resource Center

Additional support and resources may be accessed through the Academic Resource Center (ARC). Support and resources include academic counseling, accommodations, assistive technology, peer tutoring, plagiarism prevention, testing center services, and writing coaching. Visit www.webster.edu/arc (<http://www.webster.edu/arc>) or Loretto Hall 40 on the main campus for more information.

Student Success Portal

Webster University's Student Success Portal, powered by Starfish, is a communications tool to connect students with faculty members and campus support services. It allows faculty and staff members to communicate with you regarding academic achievements as well as areas where support may be helpful. You may use the portal to contact faculty and staff members for assistance and to arrange meetings. Activity in the portal will be communicated to your Webster University email account. The Student Success Portal is available via your course home page in WorldClassRoom or via Connections. Learn more about the Student Success Portal at <http://www.webster.edu/success/students.html> (<http://www.webster.edu/success/students.html>).

University Library

Webster University Library is dedicated to supporting the research needs and intellectual pursuits of students throughout the University's worldwide network. Resources include print and electronic books, journal articles, online databases, DVDs and streaming video, CDs and streaming music, datasets, and other specialized information. Services include providing materials at no cost and research help for basic questions to in-depth exploration of resources. The gateway to all of these resources and services is <http://library.webster.edu> (<http://library.webster.edu/>). For support navigating the library's resources, see <http://libanswers.webster.edu/> (<http://libanswers.webster.edu/>) for the many ways to contact library staff.

Drops and Withdrawals

Drop and withdrawal policies dictate processes for students who wish to unenroll from a course. Students must take proactive steps to unenroll; informing the instructor is not sufficient, nor is failing to attend. In the early days of the term or semester, students may DROP a course with no notation on their student record. After the DROP deadline, students may WITHDRAW from a course; in the case of a WITHDRAW, a grade of W appears on the student record. After the WITHDRAW deadline, students may not unenroll from a course. Policies and a calendar of deadlines for DROP and WITHDRAW are at:

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/enrollment.html>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/enrollment.html>)

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/enrollment.html>
(<http://www.webster.edu/catalog/current/graduate-catalog/enrollment.html>)

Academic Calendar - <http://www.webster.edu/academics/academic-calendar/>
(<http://www.webster.edu/academics/academic-calendar/>)

Current tuition rates, policies, and procedures, including details of pro-rated tuition refunds, are available in the "Tuition, Fees, and Refunds" section of Webster's Academic Catalogs:

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/tuition.html>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/tuition.html>)

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/tuition.html> (<http://www.webster.edu/catalog/current/graduate-catalog/tuition.html>)

Student Handbook and Other Important Policies

Student handbook and other non-academic policies may apply to you and may impact your experience in this class. Such policies include the student code of conduct, privacy, technology and communications, and more. Please review the handbook each year and be aware of policies that apply to you. The handbook is available at:

<http://www.webster.edu/student-handbook/> (<http://www.webster.edu/student-handbook/>)

Sexual Assault, Harassment, and Other Sexual Offenses

Webster University makes every effort to educate the community to prevent sexual assault, harassment, and other sexual offenses from occurring, and is committed to providing support to those affected when this behavior does occur. To access information and resources or to review the Policy on Sexual Assault, Harassment, and Other Sexual Offenses, visit:

<http://www.webster.edu/sexual-misconduct/> (<http://www.webster.edu/sexual-misconduct/>)

Research on Human Subjects

The Webster University Institutional Review Committee (IRB) is responsible for the review of all research on human subjects. The IRB process applies to all Webster University faculty, staff, and students and must be completed prior to any contact with human subjects. For more information on the IRB, visit:

<http://www.webster.edu/irb/index.html> (<http://www.webster.edu/irb/index.html>)

Course Evaluations

At the end of this course, you will have the opportunity to provide feedback about your experience. Your input is extremely valuable to the university, your instructor, and the department that offers this course. Please provide your honest and thoughtful evaluation, as it helps the university to provide the best experience possible for all of its students.

Important Technology Information

Connections Accounts

Webster University provides all students, faculty, and staff with a University email account through Connections. Students are expected to activate their Connections account and regularly check incoming University email. Students may choose to have their University email forwarded to an alternate email address. Connections account holders can call the Help Desk (314-246-5995 or toll free at 1-866-435-7270) for assistance with this setup. Instructions are also provided on the Information Technology website at:

<http://www.webster.edu/technology/service-desk/> (<http://www.webster.edu/technology/service-desk/>)

WorldClassRoom

WorldClassRoom is Webster's Learning Content Management System (LMS). Your instructor may use WorldClassRoom to deliver important information, to hold class activities, to communicate grades and feedback, and more. WorldClassRoom is available using your Connections ID at:

<https://worldclassroom.webster.edu/> (<https://worldclassroom.webster.edu/>)

Webster Alerts

Webster Alerts is the University's preferred emergency mass notification service, available free to current students, faculty and staff at all US campuses. By registering a valid cell phone number and email address, you will receive urgent campus text, voice mail and email communications. Valuable information concerning a range of incidents affecting you - from weather-related campus closures, class delays and cancellations, to more serious or life-threatening events - are immediately and simultaneously delivered through multiple communication channels. To register for Webster Alerts, visit:

<http://www.webster.edu/technology/services/webster-alerts/> (<http://www.webster.edu/technology/services/webster-alerts/>)